Lecture 1

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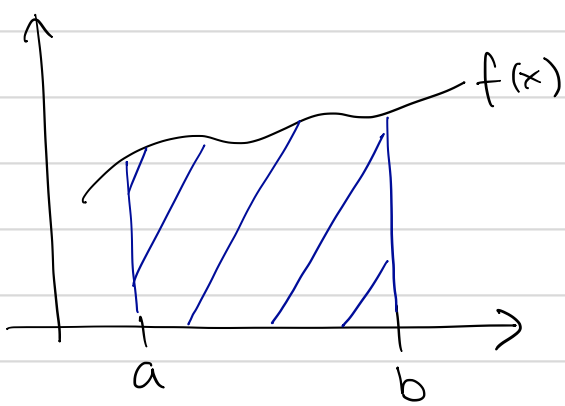
Office: PDL C-526
Hours: MTh 3:30–5:00 pm

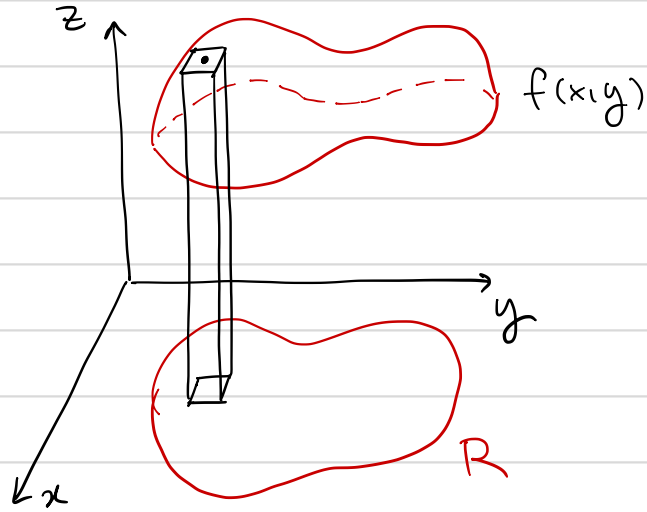
Text: Stewart, "Calculus: Early Transcendentals" (8th ed. + Webassign)

Today: Double integrals (15.1–3)

Recall:

$$\int_a^b f(x)\,dx = \text{Area below graph } y=f(x) \text{ over } [a,b]$$

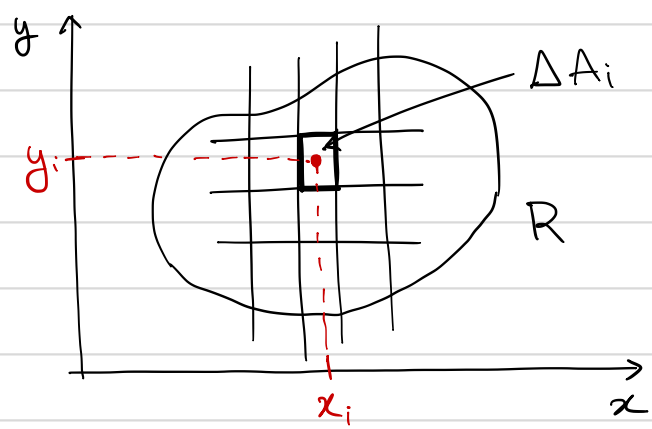




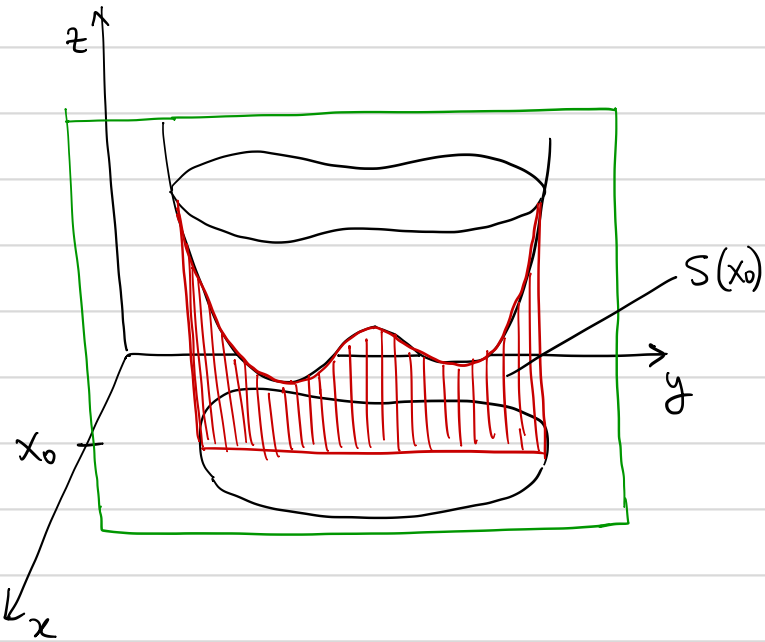
$$\iint_R f(x,y)\,dA = \text{Volume below graph } z=f(x,y) \text{ over region } R \text{ of } xy\text{-plane}$$

Riemann sums

Cut R into small pieces of area ΔA_i.

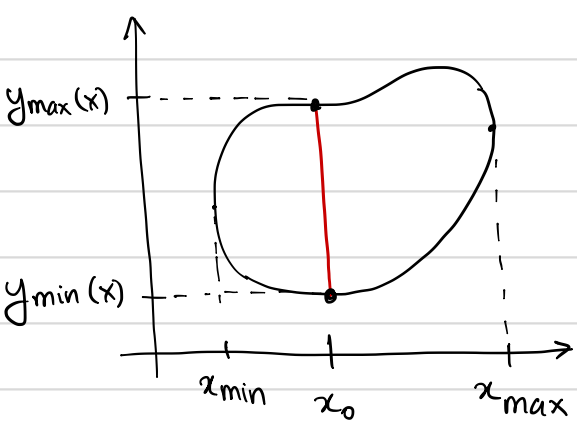


$$\boxed{\sum_i f(x_i,y_i)\,\Delta A_i \xrightarrow[\Delta A_i \to 0]{} \iint_R f(x,y)\,dA}$$



Let $S(x_0)$ = area of slice by plane // yz.

Then Volume $= \int_{x_{min}}^{x_{max}} S(x)\,dx$.



For given x,

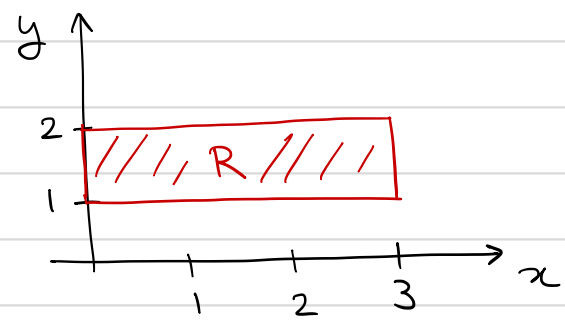
$$S(x) = \int_{y_{min}(x)}^{y_{max}(x)} f(x,y)\,dy$$

Fubini's theorem:

$$\iint_R f(x,y)\,dA = \int_{x_{min}}^{x_{max}} \left[\int_{y_{min}(x)}^{y_{max}(x)} f(x,y)\,dy \right] dx$$

Iterated integral

Example: $f(x,y) = x^2 y$



$R = \{(x,y) \mid 0 \le x \le 3,\ 1 \le y \le 2\}$

$$\iint_R x^2 y\,dA = \int_{x=0}^{3} \left[\int_{y=1}^{2} x^2 y\,dy \right] dx$$

Inner integral:

$$\int_1^2 x^2 y\,dy = x^2 \frac{y^2}{2}\Big|_{y=1}^{2}$$

$$= x^2\left(4 - \frac{1}{2}\right) = \frac{3}{2}x^2$$

Outer integral:

$$\int_0^3 \frac{3}{2}x^2\,dx = \frac{3}{2}\,\frac{x^3}{3}\Big|_0^3 = \boxed{\frac{27}{2}}$$